

PCA compression explorer

18YZALG – Basics of Algorithmization | Tutorial 10: Mini-Project

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What are we solving?

Input: A data matrix $\mathbf{X}$ with $\mathbf{N}$ samples and $\mathbf{D}$ dimensions.

Output: A compressed version of the data with fewer dimensions ($\mathbf{K} < \mathbf{D}$).

Edge Cases:

- Data with zero variance (all values identical).
- Perfectly correlated features (redundant info).

Why it's interesting: PCA allows us to visualize high-dimensional data in 2D or 3D and saves storage by removing "noise" and redundant information.

Mathematical Object

Data Representation: Data points are treated as vectors in a D-dimensional space.

Core Assumption: Features are correlated, allowing for linear redundancy reduction.

Preprocessing: Data must be centralized before processing.

Formula: $Z = X - \mu$ (Subtracting the mean of each feature)

Zero Mean Centering

The Reason Behind Centering

PCA is designed to find directions of **maximum variance**. If the data is not centered at the origin (0,0), the first principal component may incorrectly point from the origin to the mean of the data cloud, failing to capture the true variance structure.

$$X_{\text{centered}} = X - \frac{1}{n} \sum_{i=1}^n x_i$$

Geometric Effect: Centering is a translation. It shifts the entire data cloud so its center of mass lies at the origin. Crucially, it does not change the relative distances, shape, or rotation of the data points.

Solving Method: Singular Value Decomposition (SVD)

Alternative Approach: Factorize the data matrix Z directly without computing Covariance.

Formula: $Z = U * \Sigma * V^T$

Key Components:

- V columns represent the principal directions (Eigenvectors).
- Singular values in Σ are directly related to the variance.

Pros: More numerically stable, faster for large datasets, standard in modern libraries.

Solving Method: Singular Value Decomposition (SVD)

SET sample_size = 40

// Create the sample dataset

FOR i = 1 TO sample_size:

 x[i] = RANDOM_UNIFORM_REAL(from = -5, to = 5)

 y[i] = (2.5 * x[i]) + RANDOM_NORMAL_REAL(mean = 0, std_dev = 1.5)

 z[i] = (0.5 * x[i]) - (1.2 * y[i]) +
 + RANDOM_NORMAL_REAL(mean = 0, std_dev = 0.5)

SET X = MATRIX_FROM_COLUMNS(x, y, z)

// Zero mean centering

SET X_mean = COMPUTE_COLUMN_MEANS(X) // Resulting size: 1 x 3

SET X_centered = X - X_mean // Subtract X_mean from every row
of X

// Decompose the centered data matrix using Singular Value
Decomposition

// $X_{\text{centered}} = U * S * V_{\text{transpose}}$

SET (U, S, V_transpose) =
SINGULAR_VALUE_DECOMPOSITION(X_centered)

// Calculate eigenvalues and Explained Variance Ratio (EVR)

SET eigenvalues = (S squared) / (sample_size - 1)

SET total_variance = SUM(eigenvalues)

SET evr = eigenvalues / total_variance

// Isolate the top 2 Principal Components (first 2 rows of
V_transpose)

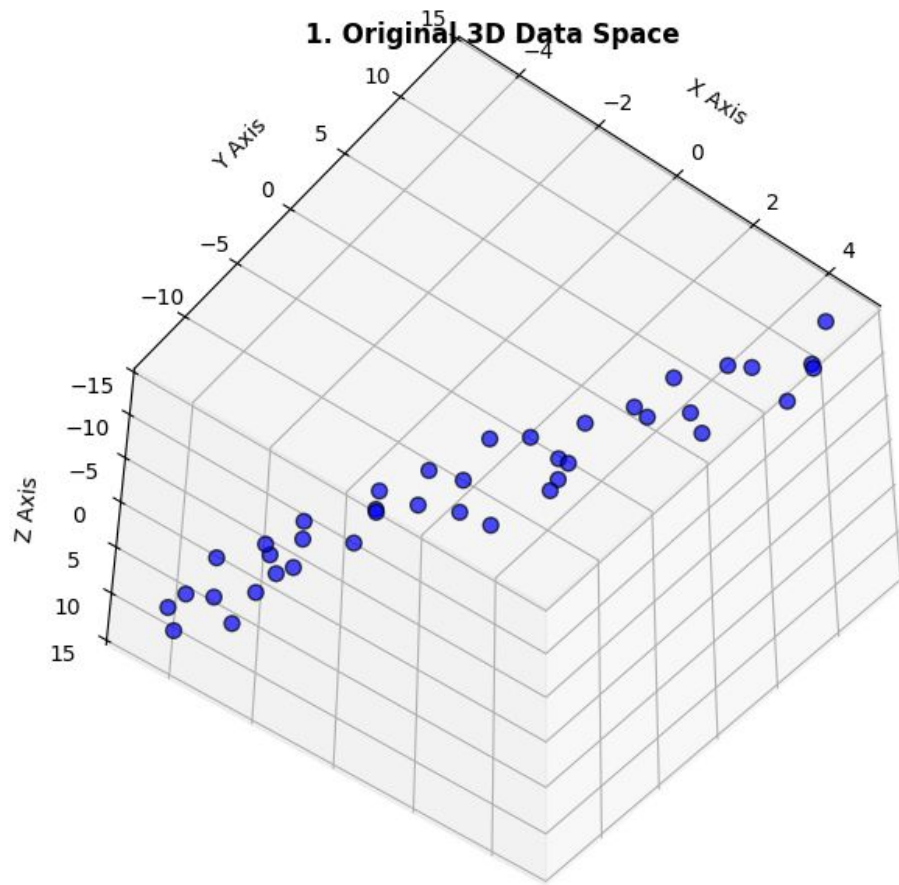
SET W2_transpose = SUB_MATRIX(V_transpose, rows = 1 to 2,
columns = all)

SET W2 = TRANSPOSE(W2_transpose)

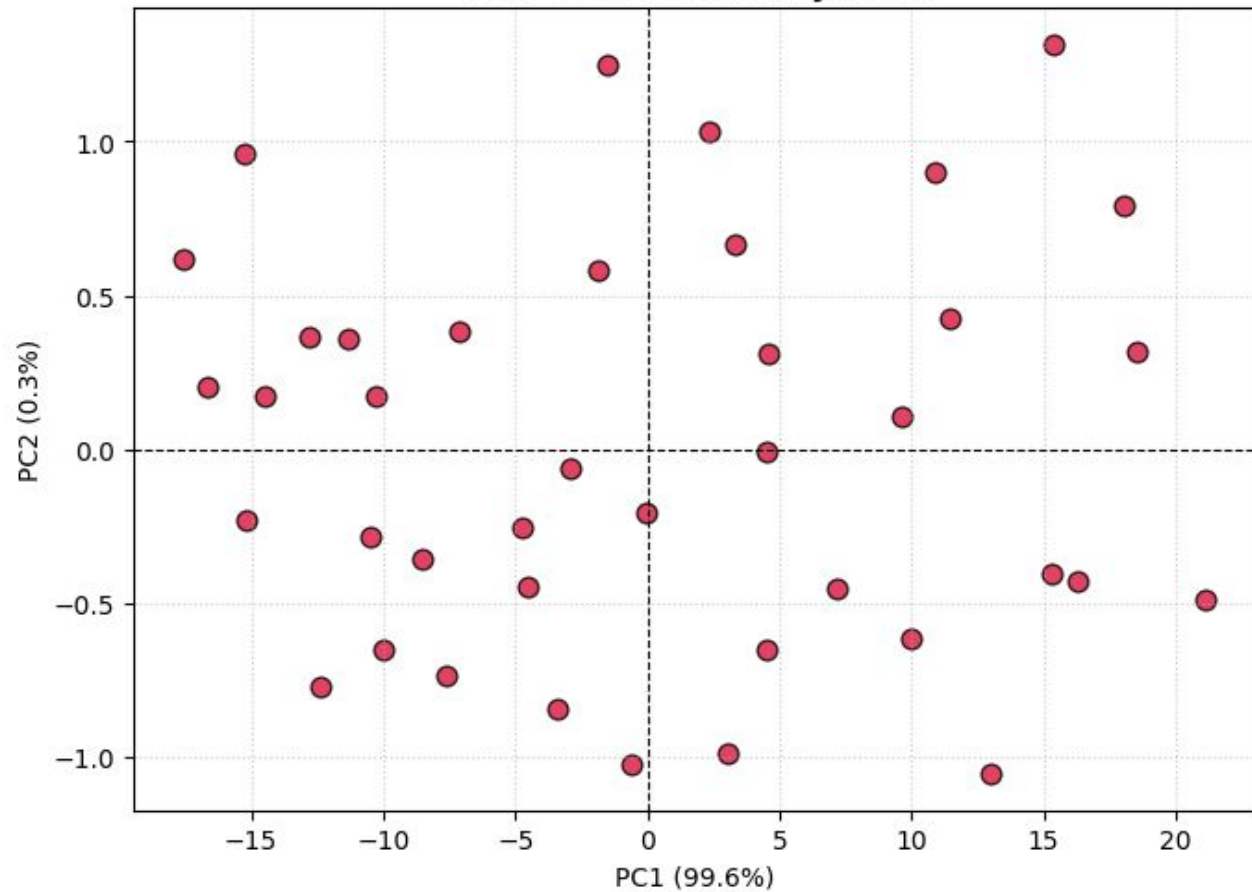
// Target transformation matrix (Size: 3 x 2)

// Project the centered 3D data down to 2D

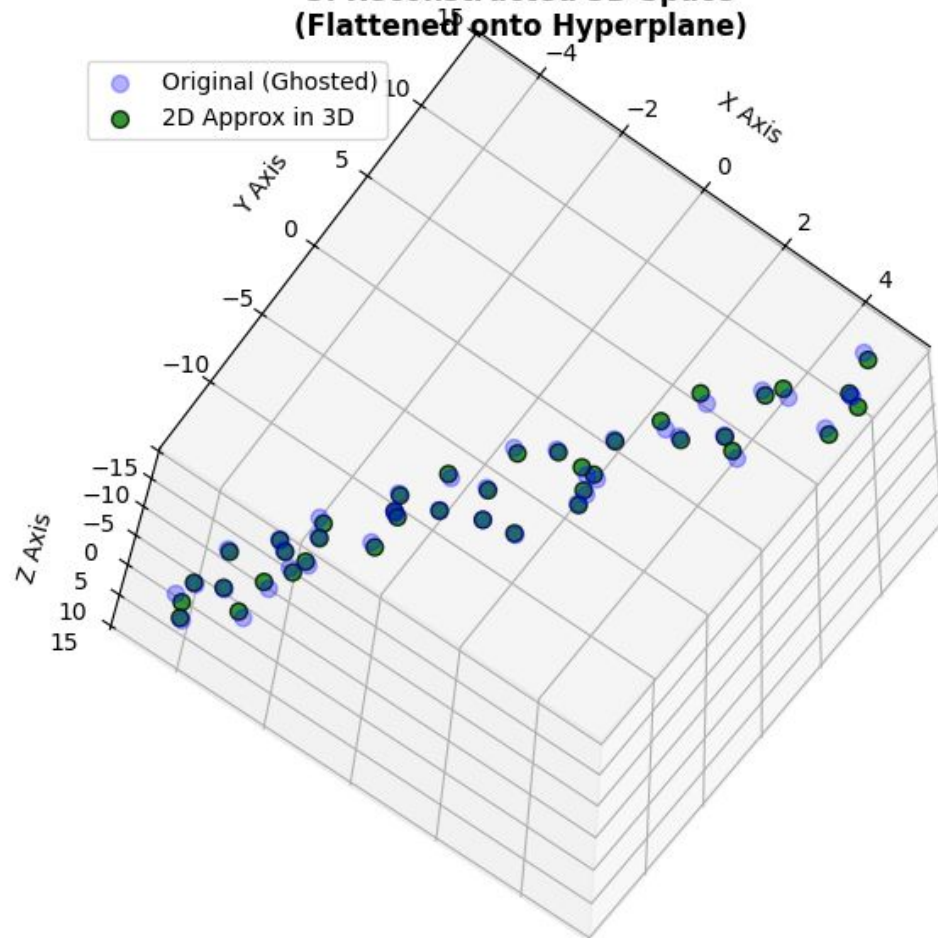
SET Z2 = MULTIPLY_MATRICES(X_centered, W2) //
Compressed coordinates (Size: N x 2)



2. Compressed 2D PCA Map (File Size Reduced by 33%)



3. Reconstructed 3D Space (Flattened onto Hyperplane)



Correctness & Assert Tests

Unit Vector Test: Principal components must have a norm equal to 1.0.

Orthogonality Check: Dot product of any two different principal components must be approximately 0.

Reconstruction Identity: If $K = D$ (no compression), the reconstructed data must match the original data within machine epsilon.

Edge Cases Handled: Matrix with zero variance, single-row inputs, and perfectly correlated features.

Explained Variance & Quality Measurement

Metric: Explained Variance Ratio measures how much information each component holds.

Example Results:

- PC1 (First Component): Captures 99.6% of total variance.
- PC2 (Second Component): Captures 0.3% of total variance.
- Cumulative Variance: Top 2 components retain 99.9% of the information.

Conclusion: We can safely drop remaining components to compress the data with minimal loss.

Performance Benchmarks

Scale Analysis: Tested with varying sizes of N (samples) and D (features).

Time Complexity: * Covariance method slows down significantly as D increases.

- SVD method scales better and uses memory more efficiently.

Reconstruction Error: Decreases exponentially as more principal components (K) are added to the model.

Conclusion & Takeaways

SVD-based PCA is the superior choice for high-dimensional engineering data.

PCA provides excellent lossy compression if features have linear correlations.

The algorithm successfully handles scientific datasets, making them ready for 2D/3D visualization.